

SAN CARLOS DE BARILOCHE, Argentina

WMO: 877650

Lat: 41.1475S

Lon: 71.1642W

Elev: 840

StdP: 91.63

Time Zone: -3.00 (W03)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	-6.9	-5.1	-11.0	1.6	1.9	-9.2	1.9	2.1	17.6	6.7	15.3	5.8	1.5	140	0.785

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	14.8	28.9	14.6	26.9	13.6	24.9	12.9	15.6	25.7	14.6	24.2	13.7	22.7	5.2	290

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
11.8	9.5	17.9	10.5	8.7	16.8	9.6	8.2	16.0	46.6	25.7	43.4	24.1	41.0	22.7	21.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 15.2	13.7	12.5	-11.2	32.3	3.2	1.4	-13.5	33.3	-15.3	34.2	-17.2	35.0	-19.5	36.0		
(5)			-11.7	17.9	3.2	1.2	-14.0	18.7	-15.9	19.5	-17.6	20.2	-19.9	21.1		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	8.6	15.1	15.0	12.3	8.6	5.8	3.3	2.6	3.6	5.6	8.2	10.7	13.2		
	DBStd	5.27	3.17	3.40	3.09	2.93	2.86	2.94	2.95	2.64	2.66	2.76	2.94	3.22		
	HDD10.0	1074	2	2	12	60	133	200	229	198	132	71	26	8		
	HDD18.3	3559	108	103	189	291	389	450	488	456	381	314	230	161		
	CDD10.0	579	161	142	83	19	3	1	0	1	2	14	46	109		
	CDD18.3	22	8	9	2	0	0	0	0	0	0	0	0	3		
	CDH23.3	746	266	253	66	2	0	0	0	0	0	4	29	127		
	CDH26.7	181	69	72	12	0	0	0	0	0	0	1	3	25		
(14) Wind	WSAvg	5.7	6.7	5.7	5.3	5.1	4.4	5.0	4.9	5.0	5.3	6.2	7.1	7.2		
(15) Precipitation	PrecAvg	791	23	21	29	59	122	153	125	115	57	43	29	29		
	PrecMax	1138	106	84	129	174	344	400	242	307	164	182	113	190		
	PrecMin	506	0	0	1	0	7	5	15	16	3	0	0	0		
	PrecStd	160	25	21	25	42	85	86	57	66	37	41	26	33		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	31.2	31.7	28.1	22.2	16.1	12.4	12.0	14.9	18.8	23.1	26.8	29.2		
		MCWB	15.2	15.7	15.1	11.9	9.8	7.8	6.9	7.9	8.7	10.4	12.5	14.6		
	2%	DB	28.8	28.9	25.1	19.5	13.8	10.2	9.9	11.9	15.9	19.9	23.3	26.7		
		MCWB	14.5	14.5	13.5	10.7	9.0	6.9	6.2	6.8	7.4	9.2	11.3	13.1		
	5%	DB	26.4	26.5	22.5	16.9	12.1	9.1	8.3	9.9	13.8	17.2	21.0	24.2		
		MCWB	13.5	13.8	12.5	9.7	8.2	6.1	5.3	5.8	6.6	8.4	10.3	12.3		
	10%	DB	24.1	24.0	20.1	14.8	10.8	7.9	7.1	8.2	11.6	14.9	18.2	21.8		
		MCWB	12.7	12.9	11.5	9.2	7.5	5.4	4.5	5.0	5.8	7.7	9.3	11.5		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	17.2	16.8	16.3	13.1	11.2	9.1	8.6	9.0	9.6	12.0	13.2	15.6		
		MCDB	27.3	27.5	25.9	19.5	14.0	11.2	10.2	12.5	16.6	19.4	24.0	26.5		
	2%	WB	15.5	15.6	14.4	11.8	10.0	7.7	6.8	7.7	8.3	10.4	12.0	14.1		
		MCDB	25.4	25.9	22.9	17.1	13.0	9.8	8.9	10.6	14.2	17.8	21.3	24.7		
	5%	WB	14.4	14.6	13.2	10.7	8.8	6.5	5.7	6.5	7.2	9.2	11.0	13.0		
		MCDB	24.2	24.1	20.8	15.3	11.4	8.5	7.8	9.3	12.2	15.7	19.3	22.3		
	10%	WB	13.4	13.6	12.1	9.7	7.9	5.5	4.7	5.2	6.3	8.2	9.9	11.9		
		MCDB	22.3	22.3	18.7	13.8	10.3	7.4	6.7	7.7	10.5	14.0	17.0	20.4		
(35) Mean Daily Temperature Range	5% DB	MDBR	14.8	15.1	14.1	11.8	8.8	7.1	7.8	8.5	10.8	11.8	12.9	13.6		
		MCDBR	20.6	20.7	19.5	17.5	11.5	8.9	10.2	11.8	15.9	18.1	19.3	20.2		
		MCWBR	9.2	9.3	10.0	10.1	7.8	6.5	7.2	7.6	9.1	9.5	9.5	9.3		
		MCDBR	17.9	17.9	16.8	13.7	9.6	7.4	8.4	9.7	13.0	14.8	16.2	17.5		
	5% WB	MCWBR	8.5	8.6	9.0	8.6	7.0	5.8	6.5	6.8	7.9	8.2	8.3	8.3		
(40) Clear-Sky Solar Irradiance	taub	0.298	0.304	0.294	0.289	0.281	0.271	0.267	0.284	0.296	0.298	0.298	0.298	0.302		
	taud	2.592	2.567	2.595	2.592	2.589	2.555	2.552	2.522	2.514	2.537	2.560	2.558			
	Ebn at Noon	1032	1000	968	906	846	820	853	896	949	995	1026	1033			
	Edh at Noon	102	99	88	75	64	60	65	79	93	101	104	106			
(44) All-Sky Solar Radiation	RadAvg	7.91	6.83	5.07	3.28	1.97	1.43	1.70	2.42	3.88	5.34	6.94	7.81			
	RadStd	0.38	0.33	0.29	0.26	0.26	0.20	0.20	0.26	0.42	0.40	0.45	0.58			

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	+0.86	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (2 neighbors)		N/A	N/A	N/A	+0.56	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page